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Introduction

I have an inclination to nonlinear plasma physics problems that can be treated by closure models, e.g. weak turbulence, dynamo, and eddy-damped quasi-normal Markovian (EDQNM), and numerical simulations. I have been working on space, high-energy astrophysical, and fusion plasmas described by fluid, gyrokinetic or kinetic models.

Interests: wave turbulence, turbulence closure modeling, kinetic Alfvén wave, non-resonant streaming instability, nonlinear gyrokinetics, solar wind, tokamak, cosmic ray.

My recent theoretical works mainly concentrate on kinetic Alfvén waves (KAWs), including: (i) based on nonlinear gyrokinetic theory, we investigate the nonlinear mode-coupling among multiple KAWs due to resonant three-wave interactions, derive the nonlinear mode equation describing the nonlinear dynamics and strong/weak turbulent evolution of KAWs in a low- β uniform magnetized plasma, and as a first application analyze the parametric decay instability with implication on spectral cascading; (ii) based on the quasi-particle picture and weak turbulence theory, we investigate the possible stationary power-law solutions of weak KAW turbulence using the Zakharov conformal transformation, and analyze its cascade behaviors due to energy and/or helicity cascade.

Education

Zhejiang University, Institute for Fusion Theory and Simulation, (Direct-To-)Ph.D.	2023 –2028
INAF, Osservatorio Astrofisico di Arcetri, Florence, joint-Ph.D.	2025 –2026
Zhejiang University, School of Physics & Chu Kochen Honors College, B.Sc.	2019 –2023

Publication

Stationary power-law solutions of weak kinetic-Alfvénic turbulence	2025
<u>Kexun Shen</u> , Zhiwen Cheng, Zhiyong Qiu [†] , submitted to ApJ, arXiv:2508.03478	
Nonlinear saturation of TAEs via ion induced scattering in nonuniform plasmas	2025
Zhiwen Cheng, <u>Kexun Shen</u> et al., Nuclear Fusion 65, 082002	
Resonant decay of KAWs and implication on spectral cascading	2024
<u>Kexun Shen</u> , Zhiwen Cheng, Zhiyong Qiu [†] , Physics of Plasmas, 31(4), 042309	
Effects of plasma nonuniformity on TAE nonlinear decay	2024
Zhiwen Cheng, <u>Kexun Shen</u> , Zhiyong Qiu [†] , Nuclear Fusion, 64(6), 066031	

Conference

Asia-Pacific Conference on Plasma Physics (AAPPS-DPP), Fukuoka, Japan	2025
Oral talk: Stationary power-law solutions of weak kinetic-Alfvénic turbulence	
Asia-Pacific Conference on Plasma Physics (AAPPS-DPP), Malacca, Malaysia	2024
Invited talk: Resonant decay of kinetic Alfvén waves and implication on spectral cascading	